ChArUco calibration target, ceramic

DATASHEET Rev A · 2026-10-24

Lithographed chrome pattern on a white alumina ceramic substrate. Feature accuracy two orders of magnitude better than printed boards, thermally stable, chemically inert. For metrology-grade camera calibration, high-magnification optics, and any setup where the target itself must not be the error budget.

Ordering

Part number	Size (W × H)	Squares (rows × cols)	Square / marker	Dictionary	Corners	Markers
CT-CHA-C-110085-{DICT}	110 × 85 mm	7 × 10	10 / 7.5 mm	6X6 · 5X5 · 4X4 · AT36	54	35
CT-CHA-C-150100-{DICT}	150 × 100 mm	7 × 11	12.5 / 9.5 mm		60	39
CT-CHA-C-185110-{DICT}	185 × 110 mm	7 × 12	14 / 10.5 mm		66	42
CT-CHA-C-230140-{DICT}	230 × 140 mm	8 × 14	15 / 11 mm		91	56

Start ID 0, row-major, OpenCV ≥ 4.6 placement. Larger ceramic plates (up to 400 × 400 mm) and other grids on request with NRE.

Specifications

Geometry

Feature position accuracy	±2 µm (any feature to nominal, at 20 °C)
Edge definition	≤ 1 µm
Orthogonality	±0.002°

Physical

Substrate	Alumina ceramic (Al_2O_3 ≥ 96 %), white, 3 mm
Pattern	Chrome, blue-black, lithographed; low gloss
Flatness	≤ 20 µm over the pattern area
Surface	Lapped; Ra ≤ 0.4 µm
Weight	≈ 11 kg/m² (150 × 100: ≈ 0.17 kg)
Edges	Ground, 0.5 mm chamfer
Mounting	Supplied in a padded case; use ACC-A-FIX-14 soft-jaw fixture or a vacuum/three-point kinematic mount. No holes.

Environment

Operating temperature	−50 to +200 °C
Thermal expansion	≈ 7.2 µm/(m·°C)
Chemical resistance	Inert to water, alcohols, common solvents

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Cleaning	Lens tissue + isopropyl alcohol; no abrasives

Deliverables

Inspection report	`CT-IR-<serial>: per-feature coordinates (CSV) + flatness + visual`
Files	`.opencv.json`, `.kalibr.yaml`, `.ros2.yaml`, `.halcon.descr`, `.dxf`, measured-coordinates CSV
Lead time	Made to order — 10–15 business days to dispatch
Options	`ILAC-MRA certificate issued by an accredited third-party laboratory, available as SVC-X-CERT-ILAC (+1-`

Compliance and customs

RoHS-compliant. HS 9031 80. Origin China; inspected in Germany. Fragile — ships in a hard case inside a padded carton.

Revision history

Rev	Date	Change
A	2026-10	First issue, planning values

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